Program 8a:

```
vedsk0831@DESKTOP-83FU7HB:~$ nano dynamic_memory_lab.c
vedsk0831@DESKTOP-83FU7HB:~$ gcc -g dynamic_memory_lab.c -o dyna
mic_memory_lab
./dynamic_memory_lab
Memory allocated using malloc() and calloc().
Memory resized using realloc().
calloc element 0 = 0
calloc element 1 = 0
calloc element 2 = 0
calloc element 3 = 0
calloc element 4 = 0
Memory released using free().
vedsk0831@DESKTOP-83FU7HB:~$ valgrind --leak-check=full --show-l
eak-kinds=all ./dynamic_memory_lab
==771== Memcheck, a memory error detector
==771== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward
 et al.
==771== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copy
right info
==771== Command: ./dynamic_memory_lab
==771==
Memory allocated using malloc() and calloc().
Memory resized using realloc().
calloc element 0 = 0
calloc element 1 = 0
calloc element 2 = 0
calloc element 3 = 0
calloc element 4 = 0
Memory released using free().
==771==
==771== HEAP SUMMARY:
==771==     in use at exit: 0 bytes in 0 blocks
```

```
Memory released using free().
==771==
==771== HEAP SUMMARY:
==771==     in use at exit: 0 bytes in 0 blocks
==771==   total heap usage: 4 allocs, 4 frees, 1,104 bytes alloc
ated
==771==
==771== All heap blocks were freed -- no leaks are possible
==771==
==771== For lists of detected and suppressed errors, rerun with:
 -s
==771== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 f
rom 0)
vedsk0831@DESKTOP-83FU7HB:~$
```